Computational Aided Tool for Energy Efficient Industry (MKKE 1243/SETK 4133)

Presented by:

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Expected Outcomes

By the end of the course, students should be able to:

- Introduction to Process Optimization
- What is optimization all about
- Applications of optimization in various fields
- Terminology in process optimization
- Formulating an Optimization Problem
- General procedure for solving optimization problem
- Formulation of objective function, decision variables and constraints

Reflections

What is Modeling & simulation?

What is optimisation ?

5 minutes

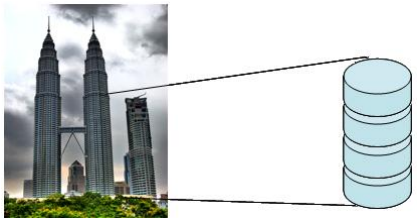
INTRODUCTION

Why Optimisation

What can happen with poor or no appropriate optimisation models?

Very Costly Lessons!

Why optimization?



- Prediction

  How many offices can be accommodated?

- Simulation

  How will the electricity demand changes during an 8-5 working hours?

- Optimization

  What type and size of solar system need to be constructed to decrease the cost of electricity at peak period

Increasing Complexity

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Optimisation Model

There is always a trade-off between time and detail

You cannot win all!

Optimisation Model

- More effort and skill can make a more realistic model

But,

- Sophisticated models are expensive

Typical Cost

Modeler	$/hr
PhD Student	40
Fresh University Graduate	120
Senior Engineer	250
Consultant	800++

What can you do with OPTIMISATION models?

Will this idea work?

Is this condition safe?

How Big do we build it?

OPTIMISATION can help answer many questions and avoid expensive mistakes

How significant is the effect?

How much can it handle?

Will I make a good ROI?

Is there a cheaper solution?

Optimization

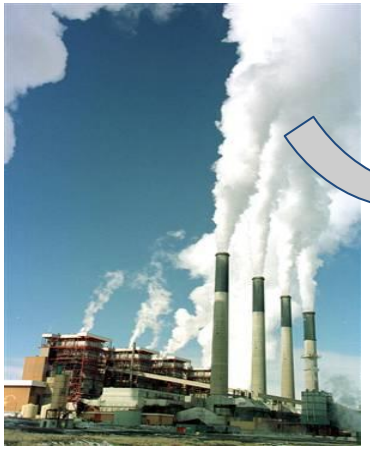
Why Optimize?

- Improved process yields
- Minimize energy consumption
- Reduced environmental pollutants
- Increase production rates
- Plan better maintenance schedule
- Improve understanding of process

Application of Optimization Models

Example: A Utility Company

GHG



Utility Plant

Heat Energy (Steam)

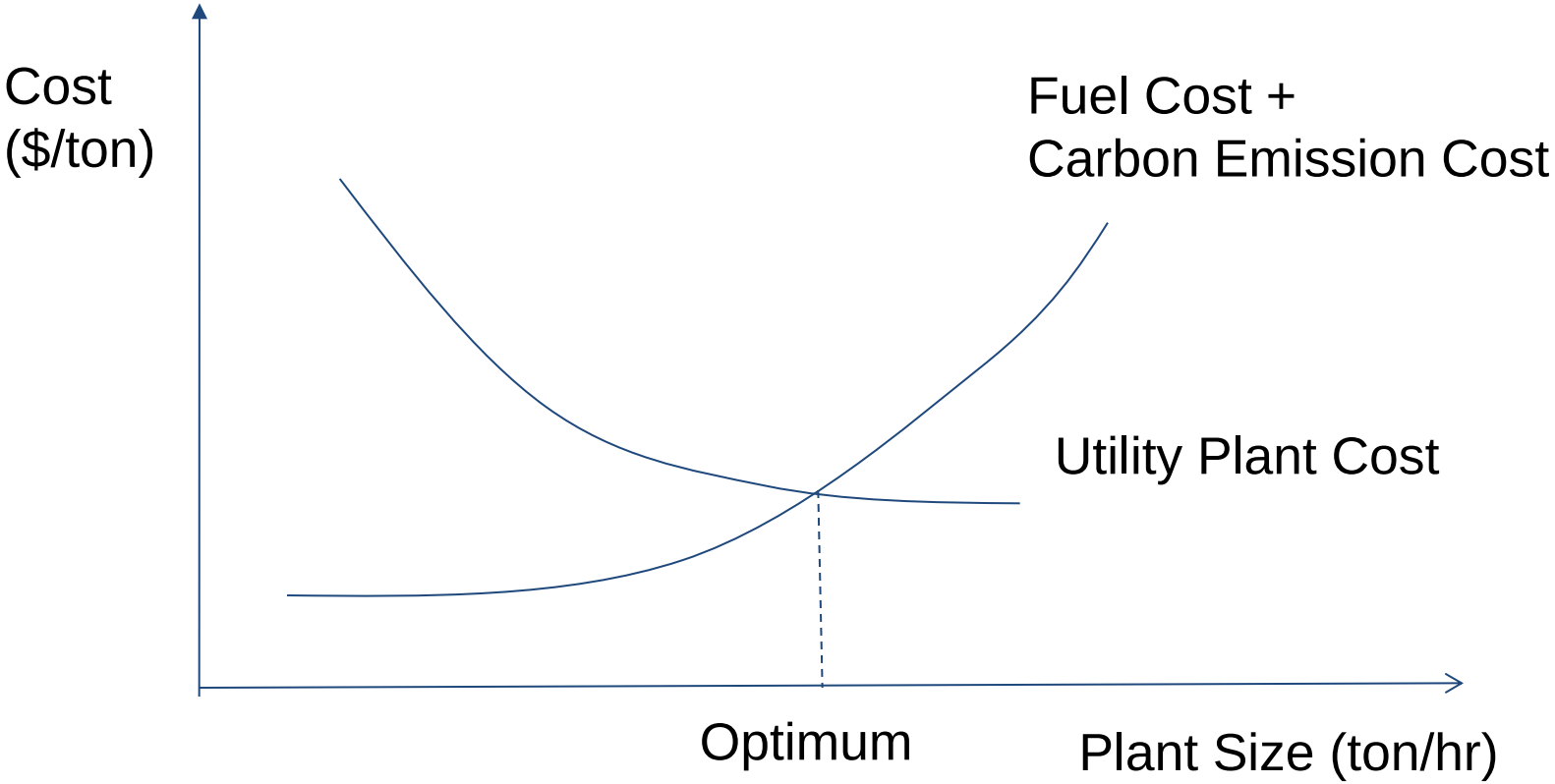
Chemical Plant

Fuel

But remember, there are always trade-offs (good/bad) to be considered

Application of Models

Example: A Utility Company



Definition of Optimisation

A mathematical technique for finding a maximum or minimum value of a function of several variables subject to a set of constraints

Finding the best solution to achieve the goal subject to a set of constraint

WHY OPTIMIZE?

Why are engineers interested in optimization?

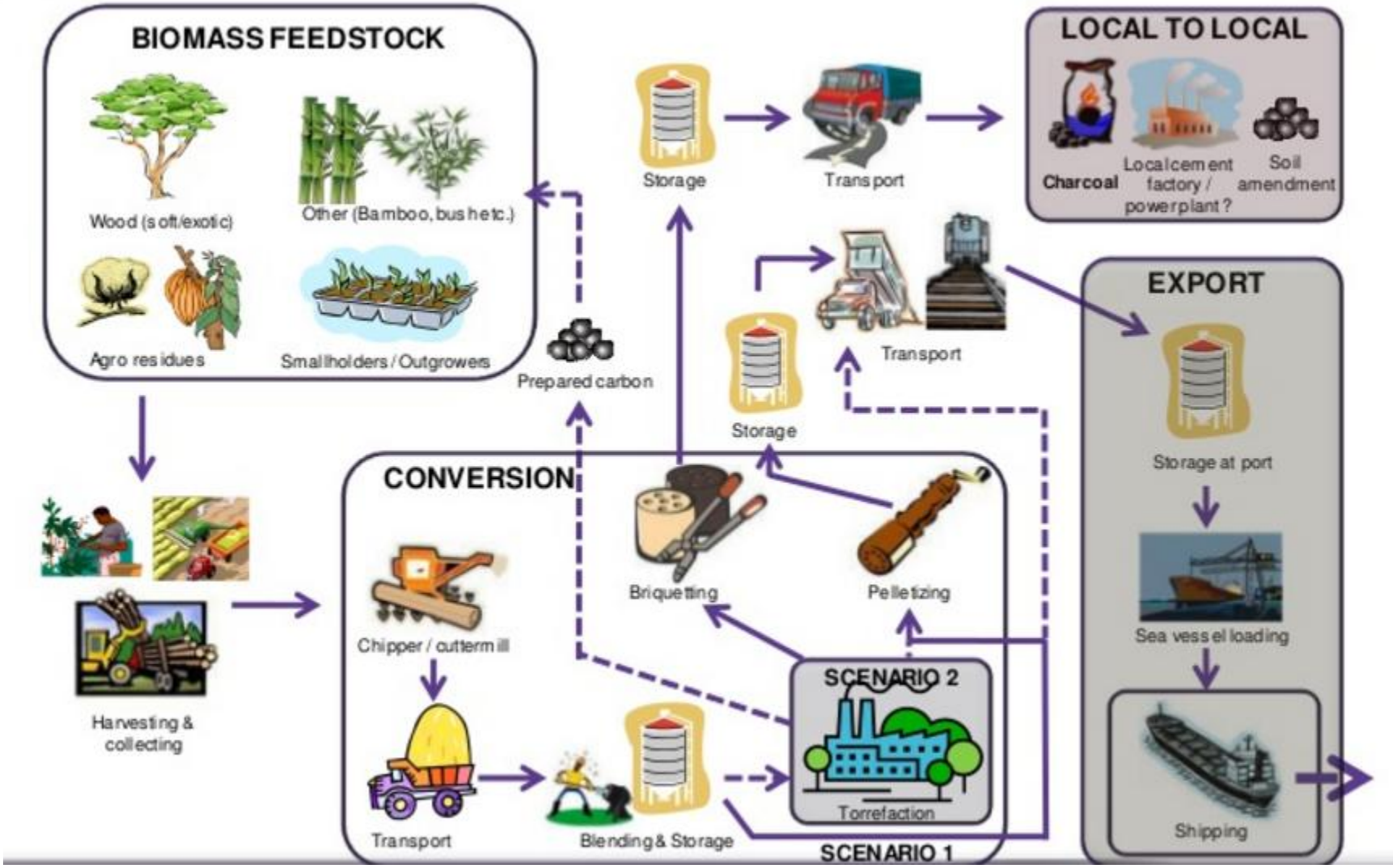
What benefits result from the optimization?

GROUP DISCUSSION 1

Please watch this video..

In your group, please discuss and identify the possible application of optimisation in energy planning

(5 minutes)

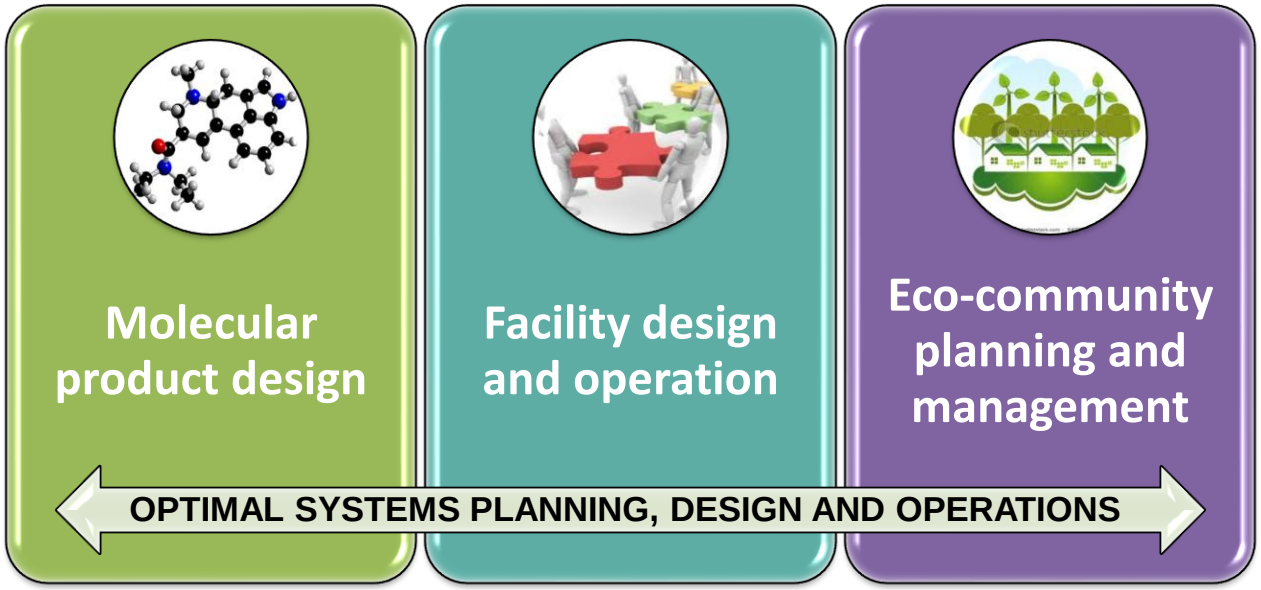

BIOMASS FEEDSTOCK
Wood (soft/exotic)
Other (Bamboo, bush etc.)
Agro residues
Smallholders / Outgrowers
Harvesting & collecting
Prepared carbon
CONVERSION
Chipper / cutter mill
Transport
Blending & Storage
Briquetting
Pelletizing
Storage
Transport
Storage
Transport
SCENARIO 2
Torrefaction
SCENARIO 1
LOCAL TO LOCAL
Charcoal
Local cement factory / power plant ?
Soil amendment
EXPORT
Storage at port
Sea vessel loading
Shipping

WHY OPTIMIZATION?

- economic incentives:
  - maximise profit
  - minimise costs
  - maximise use of resources
  - minimise effort
- technical incentives:
  - safer design and operating strategy
  - Better processing strategy
  - increased yield, conversion
  - optimum selectivity
  - minimum holdup
- integral part of design and operation:
  - optimisation of process structure (synthesis)
  - optimisation of operation
  - technical/economic
  - control

- environmental incentives:
  - Minimise carbon emission
  - Minimise environmental impact

OPTIMIZATION LEVEL

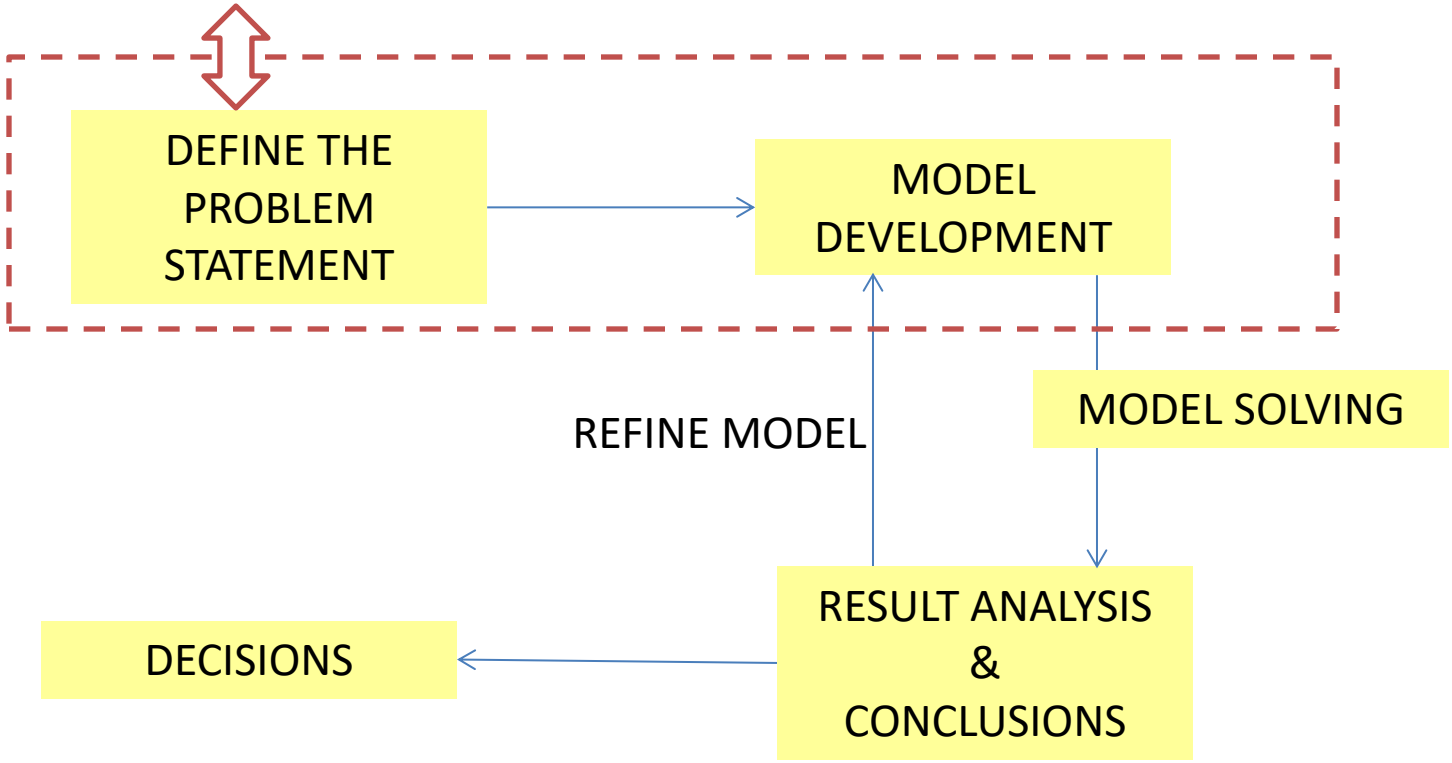


OPTIMIZATION IN ENGINEERING

The purpose of optimization is to *maximize (or minimize)* the value of a function *(called objective function)* subject to a number of restrictions *(called constraints).*

WHAT IS OPTIMIZATION ALL ABOUT

- Objective function
- Variable
- Constraints
- Parameter



Key elements of Optimisation

Objective function

– mathematical function to be minimised or maximised, related to the desired incentives by the company

Decision variables

- variables that you can adjust or choose in order to minimize or maximize the objective function
- variables can be:
  - Continuous, e.g. flowrates, concentration, temperature, pressure
  - Integers, eg. # of trays
  - Binary (either 0 or 1), eg. Select/not select - plant process flowsheet

Key elements of Optimisation

Constraints

- Limitations or restrictions of decision variables.
- can be physical, economic, policy, or environmental constraints
- can be either equality constraints or inequality constraints

Parameters

- variables that predefined by the users.
- eg. Cost of the product

EQUALITY CONSTRAINTS

Equality constraints, $h_k(x) = 0$,

- represent process restrictions:
  - Product purity must be exactly 98%

- represent the process model:
  - The heat and material balance
  - Mole fraction summation equations
  - Equilibrium equations, etc.

INEQUALITY CONSTRAINTS

Inequality constraints, $g_j(x) \geq 0$,

- represent restrictions imposed by the designer and by nature.
  - Mole fraction flowrate must be greater than or equal to zero (no negative flowrates are allowed)
  - Pressure in a particular vessel cannot exceed an upper limit
  - A reactor volume cannot exceed the maximum currently manufactured
  - Production rates and product purities must all be greater than minimum values.

EXAMPLE OF OBJECTIVE FUNCTION AND CONSTRAINTS

1. Maximize reactor conversion

   Subject to

   - reactor modeling equations
   - kinetic equations
   - limitations on T, P and $\underline{x}$

2. Minimize cost of plant

   Subject to

   - mass & energy balance equations
   - equipment modeling equations
   - environmental
   - technical and logical constraints

OPTIMISATION FOR DUMMIES

How to minimize the cost?

EXAMPLE 1: NUTRIENT REQUIREMENTS

Given a set of foods, along with the nutrient information for each food and the cost per serving of each food, the objective of the diet problem is to select the number of servings of each food to purchase (and consume) so as to minimize the cost of the food while meeting the specified nutritional requirements.

Identify the possible objective function, variable, parameter and constraints

SOLUTION TO EXAMPLE 1

Given a set of foods, along with the nutrient information for each food and the cost per serving of each food, the objective of the diet problem is to select the number of servings of each food to purchase (and consume) so as to minimize the cost of the food while meeting the specified nutritional requirements.

Objective Function:

- To minimise the cost

Variable:

- Number of servings of each food

Parameter:

- Nutrient information for each food
- Cost per serving

Constraints:

- Specified nutritional requirements

EXAMPLE 2: PRODUCTION OF TONGKAT ABU

IBD produces the drug Tongkat Abu in huge batches by heating a chemical mixture in a pressurized container. Each time a batch is produced, a different amount of Tongkat Abu is produced. The production *yield z* (kg product/ kg reactant) is driven by the volume, pressure, temperature, composition of mixture.

Following is the limitation of process:

- Volume (V) must be between 1 and 5 liters
- Pressure (P) must be between 200 and 400 milliliters
- Temperature (T) must be between 100 and 200 degrees centigrade
- Mixture must be made up entirely of A, B, and C
- For the drug to perform properly, only half the mixture at most can be product A.

Identify the possible objective function, variable, parameter and constraints

SOLUTION TO EXAMPLE 2

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Objective function

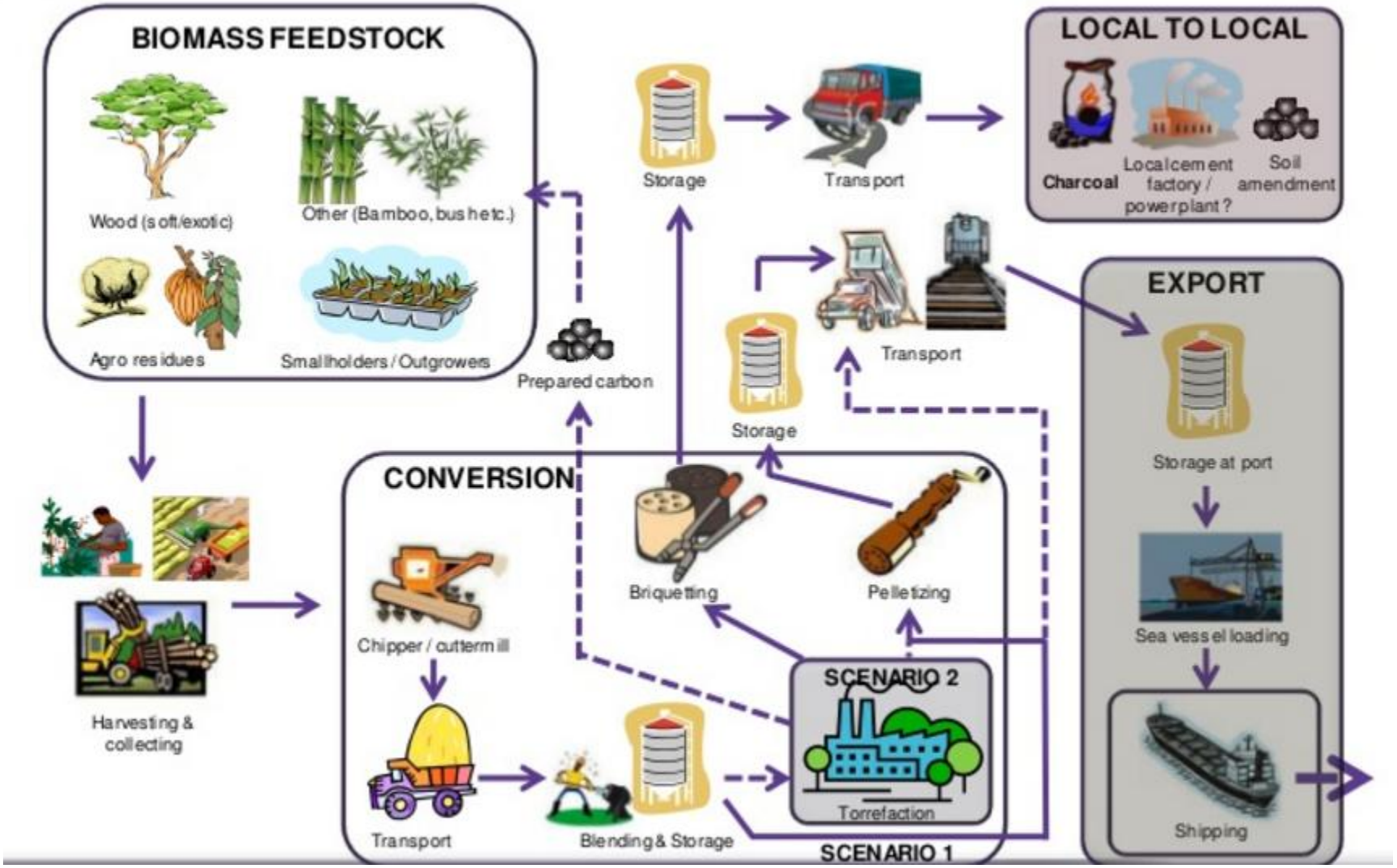
To maximise z

Variable

- V
- P
- T
- A
- B
- C

Constraints

- Volume range
- Pressure range
- Temperature range
- Presence of A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GROUP DISCUSSION 2

In your group, please formulate the possible optimisation problem statement and identify the relevant objective function, decision variable, parameter, and constraints.

(10 minutes)

FORMULATION OF OPTIMISATION MODEL

Model vs Model

vs

Objective Function

min (or max) $f(x_1, x_2, \ldots\ldots, x_N)$

subject to

Inequality Constraints

$g_1(x_1, x_2, \ldots\ldots, x_N) \leq 0$

$g_2(x_1, x_2, \ldots\ldots, x_N) \leq 0$

$g_m(x_1, x_2, \ldots\ldots, x_N) \leq 0$

Equality Constraints

$h_1(x_1, x_2, \ldots\ldots, x_N) = 0$

$h_2(x_1, x_2, \ldots\ldots, x_N) = 0$

$h_E(x_1, x_2, \ldots\ldots, x_N) = 0$

EXAMPLE 3:
FORMULATION OF OPTIMISATIO MODEL

Julia wonders if she can keep the amount of fat in her diet down and still get all the protein (45 g), carbohydrate (256 g), and calories (1,980 kcal) that she needs every day by eating fast food only. For other nutrients, such as vitamins, iron, and calcium, she will depend on pills. She chooses her favorite fast foods: hamburger and French fries. The nutrient values per serving are given below. What is the optimum servings of hamburger and fries would Julia need to eat to satisfy her daily diet requirements (minimum total amount of fats)?

	Hamburger	Fries
Fat (g)	10	18
Protein (g)	15	3
Carbohydrate (g)	32	32
Calories (kcal)	220	396

SOLUTION TO EXAMPLE 3

Julia is looking for

decision variables

that minimizes the total amount of fat

objective function

subject to the following minimum diet requirements

constraints

Protein constraint

Carbohydrate constraint

Calories constraint

nonnegativity constraints

SOLUTION TO EXAMPLE 3

Julia is looking for

decision variables

HB = the number of servings of Hamburger

FF = the number of servings of French Fries

TF = total amount of fat

that minimizes the total amount of fat

objective function

$$TF = 10\,HB + 18\,FF$$

subject to the following minimum diet requirements

constraints

$$15\,HB + 3\,FF >= 45$$ Protein constraint

$$32\,HB + 32\,FF >= 256$$ Carbohydrate constraint

$$220\,HB + 396\,FF >= 1980$$ Calories constraint

nonnegativity constraints

$$HB >= 0 \text{ and } FF >= 0$$

EXAMPLE 4

IBD produces the drug Tongkat Abu in huge batches by heating a chemical mixture in a pressurized container. Each time a batch is produced, a different amount of Tongkat Abu is produced. The production *yield z* (kg product/ kg reactant) is affected by the volume, pressure, temperature, composition of mixture.

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$$\text{Yield} = 300 + 0.8V + 0.01P + 0.06T + 0.001T*P - 0.01T^2 - 0.001P^2 + 11.7A + 9.4B + 16.4C + 19A*B + 11.4A*C - 9.6B*C$$

SOLUTION TO EXAMPLE 4

The Complete IBD Optimization Model

$$\text{maximize } z = 300 + 0.8V + 0.01P + 0.06T + 0.001T*P - 0.01T^2 - 0.001P^2 + 11.7A + 9.4B + 16.4C + 19A*B + 11.4A*C - 9.6B*C$$

Subject to (s.t.)	$V \leq 5$	$T \leq 200$	$A \geq 0$
	$V \geq 1$	$T \geq 100$	$B \geq 0$
	$P \leq 400$	$A + B + C = 1.0$	$C \geq 0$
	$P \geq 200$	$A \leq 0.5$	

Reflection

- 7.1 Formulation of the Optimisation Problem
    1. Introduction
    2. Define the problem statement
    3. Formulation of optimisation model

End of lesson

Questions ???

SAMPLE PROBLEM FORMULATION

A farmer is preparing to plant a crop and needs to fertilize a field. There are 2 brands of fertilizer to choose from, Super-gro and Crop-quick. Each brands yield a specific amount of nitrogen and phosphate per bag as follows:

	Chemical Contribution	
	Nitrogen (kg/bag)	Phosphate (kg/bag)
Super-gro	2	4
Crop-quick	4	3

The farmer requires at least 16 kg of N2 and 24 kg of phosphate. Super-gro cost $ 6/bag and Crop-quick costs $3/bag. The farmer wants to know how many bags of each brand to purchase in order to minimize the total cost of fertilizing.

SOLUTION EXAMPLE 4

STEP 1: Define the decision variables

x_1 = bags of Super-gro

x_2 = bags of Crop quick

STEP 2: Define the objective function

min $z = 6x_1 + 3x_2$

STEP 3: Define the equality and inequality constraints

- Equality constraint

  - None except if the farmer said that the phosphate requirement must be exactly 24 pound, the constraint should be written

  $4x_1 + 3x_2 = 24$ (phosphate constraint)

- Inequality constraint

  $2x_1 + 4x_2 \geq 16$ (nitrogen constraint)

  $4x_1 + 3x_2 \geq 24$ (phosphate constraint)

- Nonnegativity constraints – to indicate that negative bags of fertilizer cannot be purchased

  $x_1, x_2 \geq 0$